

Homework 4

Problem 1. Count the permutations with exactly k fixed points. (Remark: π is a permutation of the set $\{1, 2, \dots, n\}$. Call an index i with $\pi(i) = i$, a fixed point of the permutation π .)

[Note: you can use $D(m)$ (i.e., the number of permutations on m items with no fixed points) directly.]

Solution. First choose the points that are fixed. It will have $\binom{n}{k}$ possible choices.

The rest is counting the number of permutation without a fixed point, which is $D(n - k)$.

In all, the answer is $\binom{n}{k} \cdot D(n - k)$. $\square$

Problem 2. What is wrong with the following inductive ‘proof’ that $D(n) = (n - 1)!$ for all $n \geq 2$? Can you find a false step in it? For $n = 2$, the formula holds, so assume $n \geq 3$. Let π be a permutation of $\{1, 2, \dots, n - 1\}$ with no fixed point. We want to extend it to a permutation π' of $\{1, 2, \dots, n\}$ with no fixed point. We choose a number $i \in \{1, 2, \dots, n - 1\}$, and we define $\pi'(n) = \pi(i)$, $\pi'(i) = n$, and $\pi'(j) = \pi(j)$ for $j \neq i, n$. This defines a permutation of $\{1, 2, \dots, n\}$, and it is easy to check that it has no fixed point. For each of the $D(n - 1) = (n - 2)!$ possible choices of π , the index i can be chosen in $n - 1$ ways. Therefore, $D(n) = (n - 2)! \cdot (n - 1) = (n - 1)!$.

Solution. Basically it says that the extended function π' is totally decided by the choice of i . However this is not the case: after letting $\pi'(n) = \pi(i)$, $\pi'(i) = n$ could not be the only choice for keeping π' bijective. In another word, the construction is an undercount. $\square$

Problem 3. How many ways are there to seat n married couples at a round table with $2n$ chairs in such a way that the couples never sit next to each other?

Solution.(hint)

$A_i = \{\text{the ways of seating in which the } i\text{th couple is adjacent.}\}$

- $|A_i| = (2n - 1)! \cdot 2^1 / (2n - 1);$

- $|A_i \cap A_j| = (2n-2)! \cdot 2^2 / (2n-2)$ for $i \neq j$;
- $\dots$;
- $|A_{i1} \cap \dots \cap A_{ik}| = (2n-k)! \cdot 2^k / (2n-k)$ for different A_{ij} s.

The final result should be $(2n)! / (2n - |A_1 \cup A_2 \cup \dots \cup A_k|)$. Then by PIE.... $\square$

Problem 4. Find generating functions for the following sequences (express them in a closed form, without infinite series!):

1. $0, 0, 0, 0, -6, 6, -6, 6, -6, \dots$.
2. $1, 0, 1, 0, 1, 0, \dots$.
3. $1, 2, 1, 4, 1, 8, \dots$

Solution.

$\square$

Sequence	Generating Function
$(1, 1, 1, 1, \dots)$	$\frac{1}{1-x}$
$(1, -1, 1, -1, \dots)$	$\frac{1}{1+x}$
$(-6, 6, -6, 6, \dots)$	$\frac{-6}{1+x}$
$(0, 0, 0, 0, -6, 6, -6, 6, \dots)$	$\frac{-6x^4}{1+x}$
$(1, 0, 1, 0, \dots)$	$\frac{\frac{1}{1-x} + \frac{1}{1+x}}{2} = \frac{1}{1-x^2}$
$(0, 1, 0, 1, \dots)$	$\frac{\frac{1}{1-x} - \frac{1}{1+x}}{2} = \frac{x}{1-x^2}$
$(1, 2, 4, 8, \dots)$	$\frac{1}{1-2x}$
$(2, 4, 8, \dots)$	$\frac{\frac{1}{1-2x} - 1}{x} = \frac{2}{1-2x}$
$(1, 0, 2, 0, 4, 0, 8, \dots)$	$\frac{1}{1-2x^2}$
$(1, 1, 2, 1, 4, 1, 8, \dots)$	$\frac{1}{1-2x^2} + \frac{x}{1-x^2}$
$(1, 2, 1, 4, 1, 8, \dots)$	$\frac{\frac{1}{1-2x^2} + \frac{x}{1-x^2} - 1}{x} = -\frac{2x^3+2x^2-2x-1}{(1-2x^2)(1-x^2)}$